

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	BTECH AI & DS	Date:	19-08-2026

Code of Conduct

/ Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. Source Sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.

2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

1. Coreference Resolution Analysis

a) Referring Expressions (Mentions) and Possible Antecedents

- **Mentions:**

- **John** (Proper noun)
- **Mary** (Proper noun)
- **the park** (Noun phrase)
- **He** (Pronoun)
- **a ball** (Noun phrase)
- **She** (Pronoun)
- **it** (Pronoun)
- **The dog** (Noun phrase)
- **him** (Pronoun)
- **they** (Pronoun)

- **Possible Antecedents:**

- For **He** / **him**: John, Mary, the park, a ball
- For **She**: John, Mary, the park, a ball
- For **it**: John, Mary, the park, a ball
- For **they**: {John, Mary}, {John, Mary, the dog}, {John, Mary, a ball, the dog}

b) Constraint Application & Resolution Process

1. **"He" (Sentence 2):**

- Gender/Number: Masculine singular → eliminates Mary (feminine), park/ball (inanimate).
- Candidate left: **John**.

2. **"She" (Sentence 3):**

- Gender/Number: Feminine singular → eliminates John (masculine), park/ball (inanimate).
- Candidate left: **Mary**.

3. **"it" (Sentence 3):**

- Gender/Number: Inanimate singular → eliminates John, Mary.
- Semantic/Animacy & Recency: Action "play with [it]" requires a playable object. "a ball" and "the park" are candidates. Recency and action suitability strongly favor **a ball**.

4. **"him" (Sentence 4):**

- Gender/Number: Masculine singular → eliminates Mary, park, ball, dog.
- Candidate left: **John**.

5. **"they" (Sentence 5):**

- Gender/Number: Plural → requires a group of entities.
- Coherence & Animacy: "went home" requires animate entities capable of traveling home together. Combines all previously introduced active animate entities: **John, Mary, and the dog**.

c) **Constraint Table**

Variable (Mention)	Initial Domain (Possible Antecedents)	Applied Constraint & Elimination	Final Resolved Link
He (S2)	{John, Mary, park}	Gender/Number: Eliminates Mary (fem.), park (inanim.)	John
She (S3)	{John, Mary, park, ball}	Gender/Number: Eliminates John (masc.), park/ball (inanim.)	Mary
it (S3)	{park, ball, John, Mary}	Gender/Number: Eliminates John, Mary. Semantic/Recency: "a ball" is more recent and semantically suitable for "play with" than "park".	a ball
him (S4)	{John, Mary, dog, ball}	Gender/Number: Eliminates Mary (fem.), dog (neut./animal), ball (inanim.)	John
they (S5)	{John, Mary, dog, ball, park}	Gender/Number: Requires plural group. Semantic: "went home" filters out inanimate objects (ball, park).	{John, Mary, the dog}

d) **Final Coreference Chains & Rewritten Paragraph**

• **Coreference Chains:**

- **Chain 1 (John):** {John, He, him}

- **Chain 2 (Mary):** {Mary, She}
- **Chain 3 (Ball):** {a ball, it}
- **Chain 4 (Dog):** {The dog}
- **Chain 5 (Group):** {John and Mary, they} → includes The dog by S5.

- **Rewritten Paragraph:**

"John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home."

e) Constraint Priority & Recency Relaxation Discussion

- **Priority Order Justification:**

1. **Gender/Number Agreement (Hard Constraint):** Grammatical mismatch produces ungrammatical or impossible interpretations (e.g., "She" cannot refer to "John" in standard contexts). It acts as a strict initial filter.
2. **Semantic Compatibility / Animacy (Hard/Semi-Hard Constraint):** Filters out pragmatically impossible connections based on verb selectional preferences (e.g., an inanimate "ball" cannot "go home").
3. **Coherence (Structural Constraint):** Ensures entities remain active across discourse units without unnatural topic shifts.
4. **Recency (Soft Preference):** Used as a tie-breaker when multiple antecedents satisfy all higher-level constraints.

- **Impact of Relaxing the Recency Constraint:**

- If recency is relaxed, distance-based preferences disappear.
- For **"it"** in S3, without recency, both "the park" and "a ball" remain grammatically valid inanimate singular antecedents. The system might incorrectly resolve "play with it" to "play with the park."
- For **"him"** in S4, if multiple masculine entities were present earlier in a longer text, relaxing recency could cause long-distance drift, picking an entity mentioned several sentences prior rather than the active subject in focus.

2. Dialog Act-Based NLP Generation

a) Discourse Planning Steps

1. **Goal Formation & Intent Framing:** Establish the core target—providing actionable advice to restore focus while delivering empathetic motivation to build confidence for tomorrow's exam.
2. **Entity & Relation Selection:** Select key entities (you, exam, concentrate) and set up an

Elaboration or **Cause-Effect** discourse relation to connect the advice (taking a break) directly to the result (improved focus).

3. **Keyword & Constraint Integration:** Select at least two required keywords (break, confident, or focus) and structure a concise 2–3 sentence structure maintaining a polite, encouraging tone without logical contradictions.

b) Candidate Responses

- **Option 1:**

Taking a short **break** right now will help **you** relax and regain your **focus** so that **you** can **concentrate** on your **exam** preparation. Because **you** have worked hard, **you** should feel **confident** in your abilities to succeed tomorrow. **You've** got this!

- **Option 2:**

If **you** take a brief **break**, **you** will clear your mind and find it much easier to **concentrate** on your **exam**. Be **confident** in what **you** already know, as a refreshed mind will help **you focus** much better tomorrow.

- **Option 3:**

You can improve your ability to **concentrate** on the **exam** by stepping away for a short **break** to refresh your mind. This simple pause will restore your **focus**, leaving **you** feeling **confident** and ready to tackle your study material.

c) Evaluation and Selection

Candidate	Constraint Satisfaction Checklist	Coherence & Quality Assessment
Option 1	• DA: Advise + Encourage • Entities: you, exam, concentrate • Keywords: break, focus, confident (3/3) • Length: 3 sentences • Relation: Cause-Effect ("so that...")	High structural coherence, but the 3rd sentence ("You've got this!") adds slight redundancy without providing new discourse value.

Option 2	• DA: Advise + Encourage • Entities: you, exam, concentrate • Keywords: break, confident, focus (3/3) • Length: 2 sentences • Relation: Cause-Effect ("If... you will...")	Strong coherence. Seamlessly bridges the action (taking a break) to the result (confidence and focus) in exactly 2 clear, encouraging sentences.
Option 3	• DA: Advise + Encourage • Entities: you, exam, concentrate • Keywords: break, focus, confident (3/3) • Length: 2 sentences • Relation: Elaboration	Clear and helpful, but reads slightly more like a general instructional statement rather than a direct, warm conversation.

- **Selected Best Response: Option 2**

- **Justification:** Option 2 best balances natural conversational tone with strict constraint satisfaction. It uses a **Cause-Effect** relation ("If you take a brief break, you will...") to directly address the user's difficulty with concentration, followed by an **Elaboration** encouraging self-belief ("Be confident..."). It reuses all entity references cleanly, integrates all three keywords naturally within 2 sentences, and provides a polished, non-contradictory blend of advice and encouragement.

d) Impact of Constraint Violations

- **Violation 1: Omitting Entity Coherence (e.g., dropping you / exam)**
 - Effect: The response becomes impersonal and generic (e.g., "Taking breaks improves focus for tests"). Dropping entity chains breaks discourse continuity, making the AI sound like a detached textbook rather than an empathetic peer responding directly to the student's personal dilemma.
- **Violation 2: Flouting Logical Consistency (e.g., contradictory advice)**
 - Effect: Giving conflicting guidance (e.g., "You need to pull an all-nighter to concentrate, but make sure you stop studying and sleep right now") creates cognitive overload and anxiety for the user. It invalidates the "Advise" dialog act, causing the user to lose trust in the system's guidance.

3. Constraint-Based WSD and RST Parsing

a) Word Sense Disambiguation (WSD) of "bank"

- **Disambiguated Sense:** Riverbank (sloped land alongside a body of water).
- **Justification via Constraints:**
 - **Selectional Preference & Semantic Compatibility:** The verb predicate flooded(x) requires an entity x capable of being inundated with water. While a financial institution building can experience localized flooding, a riverbank (bank) situated immediately adjacent to a river is semantically selected by both the spatial modifier "by the river" and the event "flooded after the storm".
 - **Coherence & Spatial Proximity:** The prepositional phrase "by the river" establishes a direct spatial relationship (location(x, y) where y = river). Co-occurrence of river, flood, and storm resolves the lexical ambiguity to the topographical sense rather than the financial institution sense.

b) Predicate Logic Representation

- **Constants:**
 - b: bank (riverbank)
 - r: river
 - s: storm
 - a: quick action
- **Predicates:**
 - riverbank(x): x is a riverbank
 - river(x): x is a river
 - storm(x): x is a storm
 - action(x): x is an action
 - quick(x): x is quick

- location_by(x, y): x is located by y
- flooded_after(x, y): x flooded after event y
- saved_by(x, y): x was saved by y

- **Logical Formula:**

$\exists b \exists r \exists s \exists a (riverbank(b) \wedge river(r) \wedge location_by(b, r) \wedge storm(s) \wedge action(a) \wedge quick(a) \wedge flooded_after(b, s) \wedge saved_by(b, a))$

(Discourse-level contrast relation links the proposition flooded_after(b, s) with saved_by(b, a) via a CONTRAST relation operator).

c) Target Language Generation

- **Tamil Translation:**

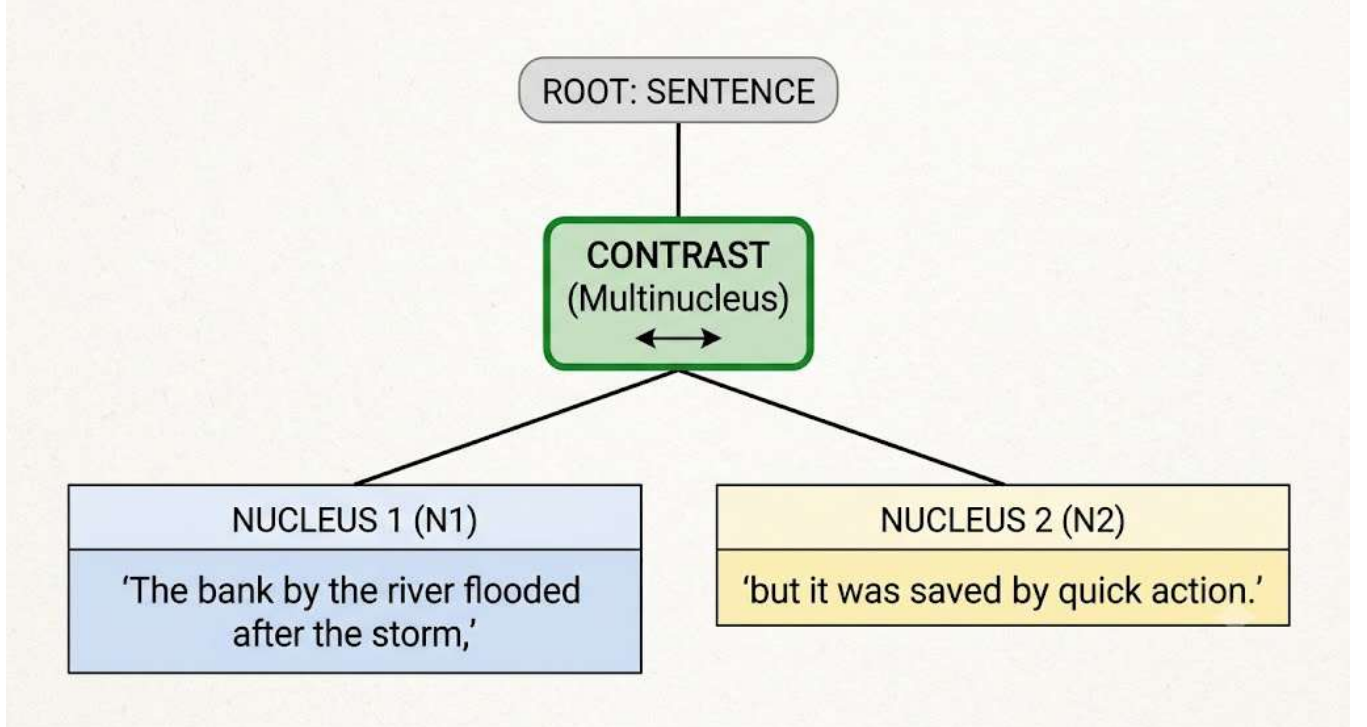
புயலுக்குப் பிறகு ஆற்றுப்பகுதி வெள்ளத்தில் மூழ்கியது, ஆனால் துரித நடவடிக்கையால் அது காப்பாற்றப்பட்டது.

- **Simple English Paraphrase:**

Although the riverbank overflowed with water after the storm hit, prompt emergency measures prevented severe damage to it.

d) Rhetorical Structure Theory (RST) Discourse Tree

The sentence contains a multinuclear **Contrast** relation linked by the discourse marker "but". Both clauses carry equal rhetorical weight in presenting opposing outcomes (flooding vs. being saved).



In standard RST conventions, multinuclear relations connect items of equal prominence without a subordinate satellite node.

e) Advantages: Constraint-Based Approach vs. Statistical Machine Translation (SMT)

1. **Deterministic Ambiguity Resolution:** SMT relies on phrase pair probabilities $P(t|s)$ and $P(s|t)$ from training corpora. If financial contexts dominate the training data, SMT frequently translates "bank" as a financial entity (e.g., வங்கி / Bank). A constraint-based approach enforces explicit semantic constraints (e.g., animacy, fluid-susceptibility, and spatial-collocation) to guarantee the correct lexical translation (ஆற்றுப்பகுதி / riverbank).
2. **Explicit Discourse Structuring:** Pure SMT translates local n-grams without explicitly preserving high-level rhetorical links. Constraint-based parsing explicitly preserves the contrastive discourse relation ($\text{Contrast}(N_1, N_2)$), ensuring that structural markers (but, however, ஆனால்) are mapped accurately to maintain clause-level coherence in the target output.
3. **Zero Information Loss & Logical Rigor:** SMT can drop modifiers or mistranslate passive constructions based on probabilistic thresholds. Constraint satisfaction validates entity-relation mappings against logic formalisms, ensuring no predicate ($\text{quick}(a)$, $\text{location_by}(b, r)$) is omitted or corrupted during generation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____